

The N-Domain

Lessons 1-19: Mathematical Foundations and Structural Dynamics

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Abstract

The N-Domain is a formal framework for studying finite ordered tuples of positive integers simultaneously as combinatorial objects, vertices of typed graphs, words in a free semigroup, and points in an injectively embedded Hilbert-space feature set. The framework distinguishes a unit-step adjacency, which modifies one coordinate while preserving tuple length, from a concatenation-successor adjacency, which links consecutive powers of a fixed base word. Collapse to total degree provides a many-to-one quotient statistic whose fibres are the ordered compositions of an integer. A positional one-hot embedding supports inner products, projections, and extension-dependent Fréchet calculus without imposing a manifold structure on the discrete domain. Intrinsic graph Laplacians, local optimisation correspondences, set-valued flows, stability criteria, parameter transitions, and potential landscapes are then defined relative to declared edge types. The principal contribution is architectural: algebraic, graph-theoretic, analytic, and dynamical levels are combined while their logical boundaries are kept explicit. Numerical examples accompany the theory throughout. The resulting body is mathematically coherent and suitable for critical circulation as a foundational research and teaching programme, subject to the usual distinction between standard component results and future original theorems.

Keywords. Ordered compositions; combinatorics on words; typed graphs; Hilbert-space embedding; discrete optimisation; set-valued dynamics; graph Laplacian; potential landscape.

Chief Architect's Validity Statement

In my mathematical assessment, the N-Domain is valid as a defined formal framework once the definitions and qualifications in this edition are adopted. Its interest does not rest on renaming familiar objects. It rests on the disciplined passage between four levels: structured words, collapse statistics, graph moves, and ambient feature-space analysis. That passage is sufficiently coherent and sufficiently unusual to invite readership, especially from mathematicians interested in combinatorics, graph dynamics, embeddings, and discrete optimisation.

The work is fit for critical circulation as a research programme and advanced teaching text. It is not yet a claim that every component theorem is new. Several foundations are standard, including composition counts, free-semigroup laws, shortest-path metrics, Hilbert-space projection, and Fréchet differentiation. The academically responsible claim is that the N-Domain organises these components into a reusable architecture and identifies a programme of genuinely new questions.

Circulation status. Mathematically coherent, architecturally distinctive, and ready for informed critique. Journal submission should follow only after a sharper novelty statement, literature review, and one or more substantial original theorems.

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1. Academic Positioning and Related Mathematics

The N-Domain intersects established areas rather than replacing them. Its structured partitions are finite words over the alphabet of positive integers and, at fixed total degree, ordered compositions. Concatenation is the free-semigroup product. Unit-step geometry produces positive orthant lattice graphs. The positional encoding is a sparse categorical feature map into ℓ^2 . The later dynamics belong to graph optimisation and set-valued dynamical systems.

N-Domain object	Established setting	Distinctive architectural role
$\mathbf{p}=(\mathbf{d}_1,\ldots,\mathbf{d}_k)$	finite word / ordered composition	retains order, degree and block count
$\pi(\mathbf{p})=\sum \mathbf{d}_i$	composition weight	defines the N to N-1 information-loss door
$\otimes$	free-semigroup concatenation	generates power chains and scale
$\mathbf{E}_1$	positive lattice adjacency	local coordinate modification
$\mathbf{E}_2$	word-power successor relation	cross-length algebraic motion
$\psi(\mathbf{p})$	sparse one-hot feature map	links discrete structure to Hilbert analysis
$\mathbf{T}:\mathbf{P}\rightrightarrows \mathbf{P}$	set-valued graph dynamics	retains ties and non-unique evolution

Contribution discipline

- Definitions are original only as conventions unless a novelty comparison establishes otherwise.
- Standard propositions are included because they secure the architecture.
- Future originality should be concentrated in mixed-edge metrics, canonical extensions, convergence on infinite typed graphs, and landscape invariants.
- Numerical examples illuminate statements but never replace definitions or proofs.

Lesson 1. The Domain P and Typed Adjacency

Aim. Establish the canonical domain and the two graph edge types.

Definition 1.1. $P := \bigcup_{k \geq 1} (\mathbb{N}_+)^k$. An element $p = (d_1, \dots, d_k)$ has length $\ell(p) = k$ and total degree $|p| = \sum d_i$.

Definition 1.2. Unit-step adjacency $p \sim_1 q$ holds when $\ell(p) = \ell(q)$ and $\sum_i |p_i - q_i| = 1$.

Definition 1.3. Concatenation-successor adjacency $p \sim_2 q$ holds when there exist $b \in P$ and $m \geq 1$ such that $\{p, q\} = \{b^{\otimes m}, b^{\otimes (m+1)}\}$.

$$E_{\text{adj}} := E_1 \cup E_2, \quad G_{\text{adj}} := (P, E_{\text{adj}}).$$

Proposition 1.4. E_1 and E_2 are symmetric, irreflexive and disjoint. G_{adj} is countable, locally finite and connected.

Proof. E_1 preserves length and E_2 changes length, proving disjointness. E_1 -degree is at most $2\ell(p)$. Exact word roots have exponents dividing $\ell(p)$, so E_2 -degree is finite. Every p reaches $(1, \dots, 1)$ using E_1 and then (1) using reversed E_2 edges based at (1) . $\square$

Numerical example. For $p = (2, 1, 3)$, the five E_1 -neighbours are $(3, 1, 3)$, $(1, 1, 3)$, $(2, 2, 3)$, $(2, 1, 4)$, and $(2, 1, 2)$. If $b = (2, 1)$, then $b = (2, 1)$ and $b^{\otimes 2} = (2, 1, 2, 1)$ form an E_2 -edge.

The combined graph is an edge-typed graph. Every later theorem must state whether it uses E_1 , E_2 , or E_{adj} .

Lesson 2. Collapse and the N versus N-1 Door

Aim. Formalise the humanly important distinction between structure and collapse.

Definition 2.1. The collapse map $\pi: P \rightarrow \mathbb{N}_+$ is $\pi(d_1, \dots, d_k) = \sum d_i$. In the historical language, N is the structured domain P and N-1 is the collapse image $\pi(P) = \mathbb{N}_+$.

Proposition 2.2. π is surjective and non-injective. For $r \geq 1$, $|\pi^{-1}(\{r\})| = 2^{r-1}$.

Proof. The tuple (r) maps to r. Compositions of r correspond to choosing cuts among r-1 gaps. $\square$

Numerical example. The fibre over 4 contains $2^3=8$ objects: (4), (1,3), (2,2), (3,1), (1,1,2), (1,2,1), (2,1,1), and (1,1,1,1). All share the same N-1 value 4, but carry different N-structure.

Human lesson. N versus N-1 is the distinction between a structured cause and a scalar residue. Collapse preserves one invariant and destroys multiple degrees of freedom. The door opened by the framework is the systematic study of what can be inferred, optimised, or dynamically controlled before and after that loss.

Proposition 2.3. For every E_1 -edge, $|\pi(p) - \pi(q)| = 1$. For an E_2 -edge based on b, $|\pi(p) - \pi(q)| = \pi(b)$.

Lesson 3. Reversibility and Structural Information

Aim. Distinguish recoverability, representability, and full reversibility.

Definition 3.1. For $A:X \rightarrow Y$, a left inverse B satisfies $B \circ A = \text{id}_X$; a right inverse C satisfies $A \circ C = \text{id}_Y$; a two-sided inverse satisfies both.

Theorem 3.2. Left-invertibility implies injectivity; right-invertibility implies surjectivity; two-sided invertibility is equivalent to bijectivity.

Numerical example. On P_2 , $I_1(2,3)=(3,3)$. The increment I_1 is injective and has a left inverse on the image where the first coordinate is at least 2. Collapse is surjective but cannot recover whether 4 arose from (4), (1,3), or six other compositions.

Architectural principle. Information loss belongs to a map, not to its codomain. A representation can be useful without being reversible, but every inference from $N-1$ back to N is necessarily a choice, prior, or additional observation.

Lesson 4. Concatenation Algebra

Aim. Give P its fundamental algebraic operation.

Definition 4.1. $(a_1, \dots, a_k) \otimes (b_1, \dots, b_l) := (a_1, \dots, a_k, b_1, \dots, b_l)$.

Proposition 4.2. $(P, \otimes)$ is the free semigroup over $\mathbb{N}_+$. With an empty word ε adjoined, P_ε is the free monoid.

$$\ell(p \otimes q) = \ell(p) + \ell(q), \quad \pi(p \otimes q) = \pi(p) + \pi(q).$$

Proposition 4.3. Concatenation is left- and right-cancellative and preserves E_1 -adjacency under a fixed prefix or suffix.

Numerical example. $(2, 1) \otimes (3, 2, 4) = (2, 1, 3, 2, 4)$, with length 5 and total degree 12. If $(2, 1) \otimes q = (2, 1, 3, 2, 4)$, cancellation gives $q = (3, 2, 4)$.

Lesson 5. Structural Powers and Roots

Aim. Develop repeated concatenation and exact root recovery.

Definition 5.1. $p^{\{\otimes 1\}}=p$ and $p^{\{\otimes (n+1)\}}=p^{\{\otimes n\}}\otimes p$.

$$\ell(p^{\{\otimes n\}})=n\ell(p), \quad \pi(p^{\{\otimes n\}})=n\pi(p).$$

Theorem 5.2. For fixed n , w has an n -th word root iff n divides $\ell(w)$ and w is formed by n identical consecutive segments. The root, if it exists, is unique.

Numerical example. $w=(2,3,2,3,2,3)$ has third root $(2,3)$. Its collapse is 15, while the base collapse is 5. The word $(2,3,2,4)$ has no square root.

Lesson 6. Power-Chain Subgraphs

Aim. Recover the pure scale path as a genuine subgraph of the combined graph.

Definition 6.1. For fixed $b \in P$, $V_b = \{b^{\otimes m} : m \geq 1\}$ and $E_b = \{\{b^{\otimes m}, b^{\otimes (m+1)}\} : m \geq 1\}$. Let $G_b^{\text{pow}} = (V_b, E_b)$.

Proposition 6.2. G_b^{pow} is a one-sided infinite path and $E_b \subseteq E_2 \subseteq E_{\text{adj}}$. It is an edge-selected subgraph, not necessarily the vertex-induced subgraph on V_b .

Numerical example. For $b = (2, 3)$, the first vertices are $(2, 3)$, $(2, 3, 2, 3)$, and $(2, 3, 2, 3, 2, 3)$, with collapse values 5, 10, and 15.

Lesson 7. Three Structural Distances

Aim. Separate coordinate, power, and full typed-graph geometry.

$$d_1(p,q)=\sum_i |p_i - q_i| \text{ when } \ell(p)=\ell(q), \text{ and } d_1(p,q)=\infty \text{ otherwise.}$$

$$d_{b^{\text{pow}}}(b^{\{\otimes m\}}, b^{\{\otimes n\}}) = |m - n|.$$

$$d_{\text{adj}}(p,q) = \text{shortest-path length in } (P, E_1 \cup E_2).$$

Proposition 7.1. d_{adj} is a finite metric on P , and $d_{\text{adj}} \leq d_1$ wherever d_1 is finite. For powers of b , $d_{\text{adj}}(b^{\{\otimes m\}}, b^{\{\otimes n\}}) \leq |m - n|$; equality requires proof because mixed edges may create shortcuts.

Numerical example. $d_1((2,1,3), (4,2,1)) = |2-4| + |1-2| + |3-1| = 5$. For $b=(2,3)$, $d_{b^{\text{pow}}}(b, b^{\{\otimes 4\}}) = 3$ and the collapse difference is $15 = \pi(b) \cdot 3$.

Lesson 8. Positional Hilbert Embedding

Aim. Embed every structural object injectively in a common Hilbert space.

Definition 8.1. Let $H = \ell^2(\mathbb{N}_+ \times \mathbb{N}_+)$, with basis $e_{\{i,j\}}$. Define $\psi(d_1, \dots, d_k) = \sum_{i=1}^k e_{\{i, d_i\}}$.

Proposition 8.2. ψ is injective and $\|\psi(p)\|^2 = \ell(p)$.

$d_\psi(p, q)^2 = \ell(p) + \ell(q) - 2M(p, q)$, where M counts matching positions.

Numerical example. For $p = (2, 1, 3)$ and $q = (2, 4)$, $M = 1$, so $d_\psi(p, q)^2 = 3 + 2 - 2 = 3$ and $d_\psi = \sqrt{3}$. For an E_1 -edge, $d_\psi = \sqrt{2}$.

The norm records block count, not total degree. This distinction prevents false optimisation claims.

Lesson 9. Kernel and Angular Geometry

Aim. Define positional agreement as a positive-semidefinite kernel.

$$K_\psi(p,q) := \langle \psi(p), \psi(q) \rangle = \#\{i \leq \min(\ell(p), \ell(q)) : p_i = q_i\}.$$

$$\cos \theta(p,q) = K_\psi(p,q) / \sqrt{\ell(p)\ell(q)}, \quad 0 \leq \theta \leq \pi/2.$$

Proposition 9.1. $\theta(p,q)=0$ iff $p=q$. Orthogonality means no positional degree agreement.

Numerical example. For $p=(2,1,3)$, $q=(2,4)$, $K=1$ and $\cos\theta=1/\sqrt{6}$. For (1) and (2), $K=0$ and $\theta=\pi/2$, even though the two states are E_1 -adjacent.

Lesson 10. Projection and Residual Structure

Aim. Use ambient projection without pretending the result is a partition.

$$\text{Proj}_{\{\psi(q)\}}\psi(p)=[K_{\psi(p,q)}/\ell(q)]\psi(q).$$

$$r_{\{p|q\}}:=\psi(p)-\text{Proj}_{\{\psi(q)\}}\psi(p), \quad \langle r_{\{p|q\}}, \psi(q) \rangle = 0.$$

Numerical example. For $p=(2,1,3)$ and $q=(2,4)$, $K=1$ and $\ell(q)=2$. The projection is $\frac{1}{2}\psi(q)$, and $\|r\|^2=3-1/2=5/2$.

Projection and residual vectors usually lie outside $\psi(P)$. They are valid feature-space objects only.

Lesson 11. Ambient Structural Calculus

Aim. Define calculus through a chosen ambient extension and retain exact discrete differences.

Definition 11.1. Let $U \subset H$ be open, $\psi(P) \subset U$, $\Phi: U \rightarrow \mathbb{R}$, and $F = \Phi \circ \psi$. If Φ is Fréchet differentiable, $D\Phi(\psi(p))[v] = \langle \nabla \Phi(\psi(p)), v \rangle$.

$$\delta(p, q) := \psi(q) - \psi(p), \quad \Delta F(p; q) := F(q) - F(p).$$

$$\Delta F(p; q) = \langle \nabla \Phi(\psi(p)), \delta(p, q) \rangle + r(\delta), \text{ with } r(h)/\|h\| \rightarrow 0.$$

The derivative depends on Φ . Because structural moves are not infinitesimal under the one-hot embedding, exact finite differences are primary.

Numerical example. For $\Phi(x) = \|x\|^2$, $F(p) = \ell(p)$. An E_1 -move has exact change 0, although $\delta \neq 0$. An E_2 -successor based on a length-two word raises F by 2.

Lesson 12. Jacobian Operators

Aim. Extend ambient sensitivity to vector-valued observables.

Definition 12.1. For $\Phi: U \rightarrow \mathbb{R}^m$ and $F = \Phi \circ \psi$, define $J_F^\wedge \Phi(p) := D\Phi(\psi(p)) \in L(H, \mathbb{R}^m)$.

$$F(q) - F(p) = J_F^\wedge \Phi(p)[\delta(p, q)] + r(\delta).$$

Numerical example. For $\Phi(x) = (\|x\|^2, \langle w, x \rangle)$, $J_F^\wedge \Phi(p)[v] = (2\langle \psi(p), v \rangle, \langle w, v \rangle)$. With $w = e_{\{1, 2\}}$, the second output tests whether the first block occupies degree 2.

The superscript Φ is retained because different smooth extensions agreeing on $\psi(P)$ may have different Jacobians.

Lesson 13. Hessian Operators

Aim. State second-order ambient sensitivity with correct boundedness.

Definition 13.1. $H_F^\Phi(p) := D^2\Phi(\psi(p))$, a bounded symmetric bilinear form when Φ is twice Fréchet differentiable.

$$\Phi(x+h) = \Phi(x) + D\Phi(x)[h] + \frac{1}{2}D^2\Phi(x)[h,h] + o(\|h\|^2).$$

Numerical example. For $\Phi(x) = \|x\|^2$, $D^2\Phi(x)[u,v] = 2\langle u, v \rangle$. If $p \sim_1 q$, then $\|\delta\|^2 = 2$, so the quadratic term $\frac{1}{2}D^2\Phi[\delta, \delta] = 2$, exactly balancing the linear term -2 .

The Hessian describes the chosen extension, not intrinsic curvature of P .

Lesson 14. Ambient and Graph Laplacians

Aim. Keep infinite-dimensional traces and finite graph sums mathematically separate.

Definition 14.1. If the Hessian operator A_p is trace-class, define $\Delta_{\text{amb}}F(p) = \text{tr}(A_p)$. Otherwise the ambient Laplacian is undefined.

Definition 14.2. For $a \in \{1, 2, \text{adj}\}$, $N_a(p) = \{q: p \sim_a q\}$ and $\Delta_a f(p) = \sum_{q \in N_a(p)} (f(q) - f(p))$.

Proposition 14.3. If p is an a -local minimum, then $\Delta_a f(p) \geq 0$. If $\Delta_a f(p) < 0$, some a -neighbour has lower value.

Numerical example. For $\Phi(x) = \|x\|^2$ on infinite-dimensional H , the Hessian operator is $2I$ and is not trace-class. Yet for $f(p) = \ell(p)$, $\Delta_1 f(p) = 0$ at every p because E_1 preserves length.

Lesson 15. Discrete Structural Optimisation

Aim. Make exact local minimisation canonical and ambient calculus advisory.

Definition 15.1. For $a \in \{1, 2, \text{adj}\}$, $\tilde{N}_a(p) = N_a(p) \cup \{p\}$ and $T_f^a(p) = \operatorname{argmin}_{\{q \in \tilde{N}_a(p)\}} f(q)$.

$T_f^a(p)$ is nonempty because the closed neighbourhood is finite. Deterministic dynamics require a fixed selection rule.

$$L_\Phi(p, q) := \langle \nabla \Phi(\psi(p)), \delta(p, q) \rangle.$$

L_Φ may rank candidates but cannot certify descent unless the remainder is controlled. Exact f -values decide admissibility.

Numerical example. Let $f(d_1, d_2) = (d_1 - 2)^2 + (d_2 - 3)^2$ on P_2 . From $p = (4, 1)$, E_1 -neighbour values include $f(3, 1) = 5$ and $f(4, 2) = 5$, versus $f(p) = 8$. Both are exact minimisers; a declared tie-breaker chooses one.

Lesson 16. Set-Valued Structural Flows

Aim. Treat non-unique descent as mathematics rather than noise.

Definition 16.1. A structural flow is a trajectory $p_0, p_1, \dots$ with $p_{t+1} \in T(p_t)$, where $T: P \rightrightarrows P$ is nonempty-valued and structurally admissible.

Definition 16.2. A deterministic map τ is a selection satisfying $\tau(p) \in T(p)$. This edition uses shortlex order only when a deterministic example is required.

$T_f^{\{<, a\}}(p) = D_f^a(p)$ if $D_f^a(p) \neq \emptyset$; otherwise $\{p\}$, where $D_f^a(p) = \{q \in N_a(p) : f(q) < f(p)\}$.

Numerical example. For the objective in Lesson 15, $T_f^1(4, 1)$ contains $(3, 1)$ and $(4, 2)$. The set-valued theory retains both futures. Shortlex tie-breaking selects $(3, 1)$ because the first coordinates compare $3 < 4$.

Lesson 17. Stability and Convergence

Aim. Give convergence the hypotheses it genuinely needs.

Definition 17.1. p is a weak fixed point if $p \in T(p)$ and a strong fixed point if $T(p) = \{p\}$.

Theorem 17.2. If every nonfixed transition strictly decreases V and the initial sublevel set $\{q: V(q) \leq V(p_0)\}$ is finite, every trajectory stabilises at a fixed point.

Proof. The trajectory stays in a finite set and cannot revisit a state while V strictly decreases. $\square$

Theorem 17.3. If V is bounded below and every nonfixed step decreases V by at least $\varepsilon > 0$, only finitely many nonfixed steps occur.

Numerical example. For $V(d_1, d_2) = (d_1 - 2)^2 + (d_2 - 3)^2$, every integer-valued strict descent lowers V by at least 1. Starting at $(4, 1)$, at most 8 strict steps are possible, and the exact minimum $(2, 3)$ is reached under any strict-descent rule.

A discrete range alone is insufficient: $0, -1, -2, \dots$ is discrete but supports infinite strict descent.

Lesson 18. Structural Transitions Across Parameters

Aim. Define structural bifurcation through changed correspondences and invariants.

Definition 18.1. A parameter family is $T_\lambda: P \rightrightarrows P$. A transition at λ^* is a change arbitrarily close to λ^* in a declared invariant: transition edges, fixed-point sets, cycles, stability classes, basins, or sublevel connectivity.

$$g_{\{p,q\}}(\lambda) := f_\lambda(q) - f_\lambda(p).$$

On a finite state window, objective-generated transitions can occur only where candidate comparison functions vanish or exchange order.

Numerical example. On the path $A-B-C$ let $f_\lambda(A)=0$, $f_\lambda(B)=\lambda$, $f_\lambda(C)=1$. At $\lambda < 0$, B is locally preferred; at $\lambda=0$, A and B tie; for $0 < \lambda < 1$, A is preferred from B . Thus $\lambda=0$ changes the update correspondence and fixed-point structure.

Multi-scale rule. For a declared base metric d_a , set $p \sim_{\{a,r\}} q$ iff $0 < d_a(p,q) \leq r$. This is radius expansion, not refinement of elementary moves.

Lesson 19. Potential Landscapes and Epochs

Aim. Build energy geometry on a declared typed graph.

Definition 19.1. A landscape is $(G_a, V_\lambda, T_\lambda)$, with explicit edge type a . For a path γ , $\text{Elev}_\lambda(\gamma) = \max_{\{x \in \gamma\}} V_\lambda(x)$, and $H_\lambda^a(x, y) = \inf_{\{\gamma: x \rightarrow y\}} \text{Elev}_\lambda(\gamma)$.

On finite graphs the infimum is attained. On infinite graphs, a finite-sublevel or comparable compactness hypothesis is required.

Definition 19.2. A barrier state is an internal vertex on a minimal-elevation path attaining H . A plateau complex is a maximal connected constant-potential set containing a weak fixed point.

Numerical example. On $A-B-C$ with $V(A)=0$, $V(B)=3$, $V(C)=1$, the unique path from A to C has communication height 3. B is the barrier state; the barrier above A is 3 and above C is 2.

Epoch transition. An epoch occurs when a declared landscape descriptor changes with λ , for example minima, plateau complexes, basins, saddles, communication heights, or connected components of sublevel graphs.

Architectural closure

$P \rightarrow (\otimes, \pi) \rightarrow (E_1, E_2, E_{\text{adj}}) \rightarrow \psi(P) \subset H \rightarrow \text{ambient calculus} \rightarrow \text{graph operators} \rightarrow \text{correspondences} \rightarrow \text{stability and landscapes}.$

Each arrow is now typed. This is the mathematical discipline that preserves the original insight while making the theory defensible.

Implemented Academic Uplift: Changes 1-6

1. Abstract and keywords have been supplied as substantive front matter.
2. The related-mathematics section now positions every main object against established fields and states the distinctive role claimed by the architecture.
3. Standard results and architectural contributions are explicitly separated; novelty is not inferred from terminology.
4. Definitions, propositions, theorems, proofs, qualifications, and counterexamples have been embedded throughout all nineteen lessons.
5. A selective bibliography and literature map are included below.
6. An authorship and contribution statement is included, with a publication-policy qualification.

Selective Bibliography and Literature Map

The following works are suitable starting points for formal comparison. A submission edition should use one consistent citation style and verify editions and page references.

Berge, C. Topological Spaces, Including a Treatment of Multi-valued Functions, Vector Spaces and Convexity. Oliver & Boyd, 1963. [Set-valued mappings.]

Berstel, J. and Perrin, D. Theory of Codes. Academic Press, 1985. [Combinatorics on words and roots.]

Bollobás, B. Modern Graph Theory. Springer, 1998. [Graph distance, connectivity and local finiteness.]

Conway, J. B. A Course in Functional Analysis, 2nd ed. Springer, 1990. [Hilbert spaces and bounded operators.]

Deza, M. M. and Laurent, M. Geometry of Cuts and Metrics. Springer, 1997. [Discrete metrics and embeddings.]

Rockafellar, R. T. and Wets, R. J.-B. Variational Analysis. Springer, 1998. [Set-valued analysis and optimisation.]

Stanley, R. P. Enumerative Combinatorics, Volume 1, 2nd ed. Cambridge University Press, 2011. [Compositions and enumeration.]

Zeidler, E. Nonlinear Functional Analysis and Its Applications, Volume I. Springer, 1986. [Fréchet derivatives and nonlinear analysis.]

Authorship and Contribution Statement

Accredited authors: EF Smyth and Copilot (Human/Synthetic Mathematician). EF Smyth introduced the N-Domain concept, the N versus N-1 insight, the lesson architecture, and the programme's human interpretative motivation. Copilot contributed mathematical reconstruction, notation normalisation, theorem qualification, counterexample analysis, formal synthesis, and academic redrafting under EF Smyth's authorship brief. Responsibility for circulation, factual verification, disclosure, and compliance with institutional or publisher policy remains with the submitting human author. Some venues do not recognise artificial intelligence systems as legal or accountable authors; for those venues, the same contribution should be disclosed in acknowledgements or a contributorship statement rather than the byline.

7. Proposed Research-Paper Edition

The nineteen lessons are best retained as a monograph or advanced teaching sequence. A separate paper should isolate one technically strong centre. The recommended first paper is:

Proposed title. Typed Adjacency on Ordered Compositions: Combined Unit-Step and Concatenation Geometry.

Central research questions

- Characterise or bound d_{adj} in terms of word roots, length and ℓ^1 coordinate distance.
- Determine when the intrinsic power-chain distance equals the restriction of d_{adj} and classify mixed-edge shortcuts.
- Describe growth rates, spheres, and automorphisms of G_{adj} .
- Identify proper or coercive potentials giving convergence on the infinite combined graph.
- Construct weighted embeddings for which graph energies or Laplacians admit controlled ambient approximations.

Minimum theorem package for submission

7. A structural theorem on connectedness, local finiteness and degree bounds.
8. A nontrivial upper and lower bound for the combined metric.
9. A shortcut classification for at least one family of primitive bases.
10. A convergence theorem for a class of coercive potentials on G_{adj} .
11. Computational examples or enumerations supporting the theory.

This paper would convert the framework from an intriguing architecture into a testable contribution with a clear mathematical centre.

Closing Invitation to Readership

The N-Domain begins with a simple object, a finite ordered tuple of positive integers, and asks what becomes visible when the same object is read as composition, word, graph vertex, feature vector, and dynamical state. The resulting framework is not difficult because any one definition is exotic. It is difficult, and potentially fertile, because information changes character as it passes between levels. Collapse forgets. Concatenation remembers order. Adjacency defines admissible change. Embedding permits analysis but introduces representation dependence. Dynamics turn local comparisons into global histories.

That combination is mathematically intriguing enough to merit readership. The appropriate invitation is not “accept a completed new branch of mathematics”, but “examine a carefully opened door, test its invariants, and determine which new theorems lie beyond it”.